

ANKITA ARUN NATH

AI Student | Data Engineer & Analyst | Software Engineer

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📍 92224 Amberg, Bavaria, Germany

🌐 <https://www.linkedin.com/in/ankita-nath-30846b139>

🔗 [ML based traffic scheduling techniques for intelligent transportation system \(Wiley Research.\)](#)

🆔 <https://orcid.org/0000-0002-3205-8261?lang=en>

🐙 <https://ankita259.github.io/>

📄 Student Visa (Germany)



👤 PROFILE

M.Sc. AI student with practical experience in software development, code optimization, and quality assurance. Skilled at developing and maintaining software solutions, implementing features, and ensuring system stability through testing and bug fixing. Structured problem-solver passionate about building robust software systems and contributing to agile development teams.

🎓 EDUCATION

M.Sc. Artificial Intelligence for Industrial Application, 03/2026 – Present
[Ostbayerische Technische Hochschule Amberg-Weiden](#) 🌐 Amberg, Bavaria, Germany

- **Key Focus:** AI, Data Analytics, Process Optimization, Machine Learning, Industrial Data Systems.

B. TECH (COMPUTER SCIENCE), [Vishwakarma Govt. Engg. College](#) 🌐 08/2016 – 08/2020 | India

- GPA:1,7 and CGPA: 8,59/10

🧠 SKILLS

Software Development: Python, C++, JavaScript, HTML/CSS, Flask, Django, Code optimization, Feature implementation.

Software Quality & Testing: Bug fixing, Code refactoring, Quality assurance, Testing, System stability, Error analysis, Pytest, Jest, MBT.

Development Methods: Version control (Git), CI/CD pipelines, Agile development, Code review, Structured coding.

Tools & Infrastructure: AWS, Azure, Docker, Git, GitHub, Jenkins, CI/CD, Monitoring tools.

Technical Skills: System architecture, Technical systems understanding, Software architecture analysis, Process optimization.

📁 WORK EXPERIENCE

Senior Software Engineer, [Synopsis India Ltd.](#) 🌐 11/2022 – 03/2026 | Bengaluru, India

- Developed and maintained software solutions for 5,000+ user platform ensuring stability and performance.
- Implemented new features and functions integrating them into existing systems.
- Performed systematic code optimization and refactoring improving maintainability by 25%.
- Analyzed and fixed errors systematically ensuring software quality and reliability.
- Supported software quality assurance through testing and validation procedures.
- Participated in agile development environment contributing ideas and improvements.
- Collaborated with teams on feature development and system integration projects.

Tech Stack: Python, SQL, Pandas, NumPy, Scikit-learn, PyTorch, AWS (S3, Lambda, EC2), Git, Docker.

- Developed and maintained software solutions modernizing legacy systems to cloud-native architecture.
- Implemented new features and integrated them into production systems.
- Performed code optimization and refactoring improving system efficiency by 15%.
- Analyzed and fixed software errors ensuring quality and system stability.
- Participated in testing and validation procedures assuring software quality.
- Contributed ideas and improvements in agile development environment.

Tech Stack: Python, Java, React.js, Flask, HTML/CSS, JavaScript, Git, Jenkins, Jira, Docker.

ACADEMIC PROJECT

LIGHT-UP: Energy-Aware AI Vision & KPI Analytics Platform

04/2026

- Developing modular software solutions integrating multiple ML models and optimization strategies.
- Implementing new features and functions supporting AI pipeline functionality.
- Performing code optimization ensuring efficient system performance and maintainability.
- Analyzing system behavior and fixing issues ensuring software stability.
- Supporting quality assurance through testing and validation procedures.
- Participating in structured development implementing improvements iteratively.

Tech Stack: Python, PyTorch, Scikit-learn, Pandas, NumPy, FastAPI, Streamlit, Git.

ML Based Traffic Police Scheduling

07/2019 – 04/2021

- Published Wiley research on [Machine learning-based traffic scheduling techniques for intelligent transportation system: Opportunities and challenges](#) .
- Implemented features supporting data processing and algorithm evaluation.
- Performed code optimization ensuring system efficiency and reliability.
- Analyzed system behavior and documented algorithms ensuring quality.
- Tested and validated solutions ensuring accuracy and system stability.

Tech Stack: Python, Scikit-learn, Pandas, NumPy, MATLAB, Bash, Git.

COURSES & CERTIFICATES

[Deep Learning Specialization](#) , Coursera

[Structuring ML Projects](#) , Coursera

[React - The Complete Guide](#) , Udemy

[Web App Technology\(Django\)](#) , Coursera

[Programming for Everybody](#) , Coursera

[Python Data Structures](#) , Coursera

LANGUAGES

English — IELTS Band C1, **German** — B1

INTERESTS

— Art (sketching, painting), Travel, Music